ASSIGNMENT – 1 (CO1)

STUDENT DETAILS	
Name	V. Naga Nischay
Register Number	192472255
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INTRODUCTION

Mobile communication networks provide voice and data connectivity to users distributed across a geographical area. A cellular network divides the service region into smaller coverage areas called cells. Each cell is served by a base station, and radio channels are reused in other cells that are sufficiently separated to control co-channel interference. Frequency reuse makes it possible to serve a large number of subscribers even when the operator has a limited amount of licensed spectrum.

This assignment considers a university township of 12 km² containing academic buildings, hostels, residential colonies and a main arterial road. The operator has 336 duplex voice channels after control-channel overhead. The expected subscriber population grows from 1,000 to 10,000, 20,000 and eventually 100,000 or more. Each subscriber generates an average busy-hour traffic of 0.025 Erlang, and the target grade of service is a call-blocking probability not greater than 2%.

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TASK 1 – DESIGN ALTERNATIVE FREQUENCY-PLANNING STRATEGIES

1.1 Strategy 1: Fixed Cluster Frequency Reuse (FCA)

In Fixed Channel Assignment (FCA), the service region is divided into repeating clusters containing N hexagonal cells. Each cell in a cluster receives a different group of channels, and the same channel group is reused in other clusters at a sufficient distance. The cluster size is selected to maintain the required SIR while providing a reasonable number of channels per cell.

For this township, a candidate cluster size of N = 16 divides the 336 available channels into 16 groups. Therefore, each cell receives 336/16 = 21 channels. The arrangement is predictable and relatively simple to operate. It is suitable as a baseline because the channel plan does not need to be recalculated for every call request.

Access-channel sharing

In FDMA, each call is assigned a separate frequency channel or frequency pair. In TDMA, multiple users share a carrier by transmitting in different time slots. In OFDMA, users are assigned groups of

orthogonal subcarriers and resource blocks. Control channels carry access requests, paging and signalling, while traffic channels carry the actual voice or data communication. The 336-channel planning pool is treated as the available traffic-channel resource after control-channel overhead.

Assumptions and design principle

- Cells are approximated as regular hexagons and the frequency pattern repeats consistently.
- Neighbouring cells exchange no channels under the basic fixed-assignment method.
- Traffic demand is sufficiently predictable for a fixed channel group to be assigned to each cell.
- Co-channel cells are separated by the planned reuse distance.
- The main design principle is spatial reuse: the same spectrum is reused in geographically separated cells.

Iterative procedure / pseudocode

The cluster-size search is iterative. Valid hexagonal cluster sizes follow $N = i^2 + ij + j^2$, where i and j are non-negative integers. The procedure checks candidate cluster sizes, calculates reuse ratio and SIR, and retains only candidates that satisfy the threshold.

```
INPUT: total_channels = 336, minimum_SIR = 12 dB
FOR i = 0 to limit:
   FOR j = 0 to limit:
      N = i*i + i*j + j*j
      IF N is valid and N > 0:
         Q = sqrt(3*N)
         Estimate SIR using selected path-loss exponent
         IF SIR >= minimum_SIR:
            channels_per_cell = floor(336/N)
            Estimate Erlang-B capacity
            Store N, Q, channels_per_cell and capacity
Select a feasible cluster size and assign channel groups
Repeat the cluster pattern across the hexagonal grid
OUTPUT: feasible cluster sizes and frequency plan
```

FCA deserves investigation because it provides a stable reference design, straightforward frequency coordination and predictable interference behaviour. Its limitation is that a lightly loaded cell cannot automatically use channels that remain idle in a neighbouring cell.

1.2 Strategy 2: Sectorised Frequency Reuse

Sectorisation divides a base-station coverage area into directional sectors. A common arrangement uses three sectors of 120 degrees each, while a denser directional arrangement may use six sectors of 60 degrees each. Each sector is served by a directional antenna that concentrates radio energy towards its intended coverage region. This reduces the strength of signals transmitted or received from many unwanted directions.

The same site can therefore support more effective frequency reuse because a user in one sector sees fewer strong co-channel interferers than in an omnidirectional arrangement. Sectorisation does not create additional licensed spectrum. It improves spatial isolation and can improve the usable SIR,

subject to antenna side lobes, front-to-back ratio, sector overlap and coordination between neighbouring sectors.

Access channels, assumptions and principle

Control and traffic resources are planned separately for each sector. The radio access method may use FDMA/TDMA or OFDMA depending on the system. The design assumes that antenna azimuth, mechanical or electrical tilt, sector boundaries and neighbouring-cell coordination are correctly configured. Its governing principle is spatial filtering: antenna directivity reduces the contribution of unwanted interferers.

Pseudocode

```
FOR each base-station site:
    Divide the coverage region into 3 or 6 sectors
    Assign a coordinated channel/resource group to each sector
    Configure antenna azimuth, tilt and transmit power
FOR each test-user position:
    Identify the serving sector
    Estimate wanted power using serving-antenna gain
    Estimate interfering powers using each antenna pattern
    Calculate SIR and check the 12 dB requirement
Adjust sector orientation, tilt and power where required
OUTPUT: sector plan, SIR estimates and capacity indicators
```

Sectorisation deserves investigation in high-demand areas such as lecture halls, hostels, the main road and the stadium. It can improve SIR without acquiring more spectrum. However, it increases antenna, radio-unit, installation and optimisation costs and may create additional handoff boundaries.

1.3 Strategy 3: Dynamic Channel Assignment and Borrowing (DCA)

Dynamic Channel Assignment does not permanently reserve every channel for one cell. Instead, a central controller or coordinated radio controller selects an available channel when a call arrives. A busy cell may borrow a channel from a lightly loaded neighbouring cell if the channel is free and the assignment does not violate reuse-distance, interference or handoff constraints. When the call ends, the channel is returned to the common or partially common pool.

The access-channel pool is therefore shared according to current demand. The controller checks channel occupancy, predicted interference, neighbouring-cell load and any resources reserved for handoffs or emergency signalling. The design principle is trunking efficiency and adaptive resource allocation.

Assumptions and pseudocode

- Cells exchange timely channel-occupancy and load information.
- Channel allocation decisions are made quickly enough to serve call arrivals.
- Neighbouring cells follow common interference and reuse constraints.
- Borrowed channels are released safely when the call ends.
- Some channels may be reserved for handoff or control signalling.

```
WHEN a call request arrives in cell c:
   List channels free in c and eligible neighbouring cells
   Remove channels that violate reuse or interference constraints
   Remove channels reserved for handoff/control
   Select the eligible channel with the lowest predicted interference
   IF a channel is available:
      Assign it and update occupancy records
   ELSE:
      Queue the request if queuing is supported; otherwise block it
WHEN the call ends:
   Release the channel and update the channel pool
```

DCA deserves investigation because traffic is not uniform. Hostels may be busy at night, academic blocks during the day and the stadium during events. Dynamic allocation can reduce blocking caused by unused channels being stranded in lightly loaded cells. It cannot increase the total licensed spectrum and must preserve SIR.

1.4 Preprocessing and cost responsibility

Before choosing a strategy, the operator must prepare a land-use map, building-height map, traffic-density map, candidate site list, road and access map, hexagonal grid and radio propagation survey. Busy-hour traffic must be estimated separately for academic blocks, hostels, residential areas, roads and event venues. Drive testing or walk testing is required to identify coverage holes, shadowing and unexpected interference.

The licensed operator normally bears the cost of traffic surveys, propagation measurements, radio planning, engineering design, site acquisition, equipment, installation and network optimisation. The university or township authority may support access to rooftops, poles, campus roads and land. However, the operator remains responsible for network performance, spectrum compliance and service quality.

Strategy	**Principle**	**Main advantage**	**Main limitation**
Fixed cluster reuse	Spatial reuse	Simple, predictable channel plan	Idle channels may be stranded
Sectorisation	Spatial filtering	Reduces dominant interference	More equipment and sector coordination
Dynamic assignment	Trunking and adaptive allocation	Uses spare channels in nearby cells	Needs fast coordination and interference checks

TASK 2 – IDENTIFY THE GOVERNING DESIGN PARAMETER

Each frequency-planning strategy is controlled by a parameter that has a strong influence on capacity, interference or resource utilisation. The analysis must focus on these parameters because they determine whether the design can meet the traffic and SIR requirements.

Strategy	**Governing parameter**	**Why it governs**	**Response to growth**

Fixed cluster reuse	Cluster size N and reuse ratio Q = D/R	N controls reuse distance, SIR and channels per cell	Select the smallest feasible N; add cells or reduce radius as demand rises
Sectorisation	Number of sectors and effective interferer count	More directional sectors can reduce strong interfering signals but increase boundaries and handoffs	Increase sectorisation in concentrated hotspots where antenna isolation is practical
Dynamic assignment	Available channel pool and per-cell offered traffic A	The controller can use eligible idle channels from other cells	Improve load reporting, borrowing decisions and interference coordination

2.1 Fixed cluster reuse

For FCA, the cluster size N determines how many channel groups are created. With S = 336 total channels, the nominal channels per cell are S/N. Increasing N increases the reuse distance and generally improves SIR, but it reduces the number of channels assigned to each cell. The reuse ratio is $Q = \sqrt{(3N)}$. The selected value must satisfy the SIR threshold without making per-cell capacity too small.

2.2 Sectorisation

For sectorisation, the effective number and strength of co-channel interferers are important. A directional antenna suppresses signals arriving from directions outside its main beam. The effective interference depends on the antenna radiation pattern, side lobes, sector orientation, antenna tilt and user location. Increasing the number of sectors may improve isolation, but it also increases the number of sector boundaries and the complexity of handoff planning.

2.3 Dynamic channel assignment

For DCA, the important quantity is the offered traffic A relative to the number of eligible channels. A cell with high offered traffic and few free channels has a greater blocking risk. A lightly loaded neighbouring cell may have spare resources, but those channels can only be borrowed if the reuse and SIR checks are satisfied. As the subscriber population increases, the controller must use more accurate load information and coordinate more frequently.

For 1,000 subscribers the township-wide offered traffic is 25 Erlangs. At 10,000, 20,000 and 100,000 subscribers it becomes 250, 500 and 2,500 Erlangs respectively. The network must therefore evolve from a coverage-led macro layout to a capacity-led hierarchical layout.

TASK 3 – DERIVE CAPACITY AND INTERFERENCE FUNCTIONS MATHEMATICALLY

3.1 Hexagonal geometry and cluster size

A regular hexagonal cell has six neighbouring cells. A cluster contains N cells, and the cluster size is determined by two integer shift parameters i and j. Moving i cells along one lattice direction and j cells along a direction separated by 60 degrees gives:

$$\mathbf{N = i^2 + ij + j^2}$$

Examples are N = 3 for (i,j) = (1,1), N = 4 for (2,0), N = 7 for (2,1), N = 12 for (2,2), and N = 16 for (4,0). These are valid cluster sizes on a regular hexagonal grid.

3.2 Reuse distance and reuse ratio

Let R be the cell radius measured from the cell centre to a vertex. The centre-to-centre distance between co-channel cells is:

$$\mathbf{D = R\sqrt{(3N)}}$$

Dividing both sides by R gives the reuse ratio:

$$\mathbf{Q = D/R = \sqrt{(3N)}}$$

For N = 16, Q = √48 = 6.928. Therefore D = 6.928R. A larger cluster size separates co-channel cells farther apart and generally reduces co-channel interference.

3.3 Hexagonal cell area and number of cells

A regular hexagon can be divided into six equilateral triangles of side R. The area of one triangle is $(\sqrt{3}/4)R^2$. Therefore:

$$\mathbf{A_cell = 6 \times (\sqrt{3}/4)R^2 = (3\sqrt{3}/2)R^2 \approx 2.598R^2}$$

The approximate number of cells required to cover a total area A is:

$$\mathbf{C = ceil(A/A_cell)}$$

For R = 0.5 km, $A_cell = 2.598 \times 0.5^2 = 0.6495$ km². For a 12 km² township, C = ceil(12/0.6495) = 19 cells. This is a geometric coverage estimate only. Real planning needs overlap, edge coverage, reserve capacity and additional cells for traffic hotspots.

3.4 Channels per cell

If S = 336 channels are divided equally among N cells in a cluster, the nominal channel count per cell is:

$$\mathbf{m = S/N}$$

For N = 16, m = 336/16 = 21 channels per cell. For cluster sizes where S/N is not an integer, practical allocation must use whole channels and the total allocation must not exceed 336.

Cluster size N	Nominal channels per cell (336/N)	Reuse ratio Q = √(3N)
3	112	3.000
4	84	3.464
7	48	4.583
12	28	6.000
16	21	6.928

3.5 Erlang-B traffic capacity

Erlang-B is suitable for a loss system in which an arriving call is blocked when all traffic channels are occupied. It assumes no waiting queue and uses the offered traffic A and number of channels m to calculate blocking probability. The recursive formula is:

B(0,A) = 1

B(m,A) = [A × B(m−1,A)] / [m + A × B(m−1,A)]

For 21 channels and a target blocking probability of 0.02, repeated evaluation gives a maximum offered load of approximately 14.036 Erlangs per cell. This is the planning traffic limit used in the remaining calculations.

3.6 Township traffic and required cells

Busy-hour offered traffic is calculated by multiplying the number of subscribers by the average traffic per subscriber:

A_total = Number of subscribers × 0.025 Erlang

Subscribers	Total offered traffic	Cells at 14.036 E/cell	Interpretation
1,000	25 E	2	Coverage and minimum spacing may dominate
10,000	250 E	18	Capacity begins to require additional cells
20,000	500 E	36	More capacity sites are required
100,000	2,500 E	179	Dense small-cell or capacity-enhancement layer is required

For 100,000 subscribers, A_total = 100,000 × 0.025 = 2,500 Erlangs. Required cells = ceil(2,500/14.036) = 179. If 179 equal hexagons cover 12 km², the approximate radius is:

R ≈ √[12/(179 × 2.598)] ≈ 0.161 km

This is a first-order estimate and excludes overlap, uneven demand, reserve capacity and real site constraints.

3.7 Co-channel SIR derivation for fixed reuse

Assume a mobile is at distance R from its serving base station and six first-tier co-channel base stations are each approximately D away. Let the received power vary with distance raised to the negative path-loss exponent n. The wanted power is proportional to R^(−n), while each interfering power is proportional to D^(−n). The total interference from six equal interferers is proportional to 6D^(−n).

SIR ≈ R^(−n) / [6D^(−n)] = (D/R)^n / 6

Since D/R = √(3N), the expression becomes:

SIR ≈ (3N)^(n/2) / 6

In decibels, SIR_dB = 10 log10(SIR). For N = 16, 3N = 48. At n = 2.5, SIR = 48^1.25/6 ≈ 21.06, giving 13.23 dB. At n = 3, SIR ≈ 55.43, giving 17.44 dB. At n = 4, SIR = 48²/6 = 384, giving 25.84 dB. The n = 2.5 open-campus case is the conservative case and has only about 1.23 dB margin over the selected 12 dB threshold.

3.8 SIR under sectorisation

Sectorisation changes the interference denominator because not all co-channel base stations contribute equally to a directional user. A more general expression is:

SIR_sector = P_wanted / Σ(P_interferer,k × G_k)

Here G_k represents the effective antenna gain in the direction of interferer k relative to the wanted direction. A sector antenna can reduce the weighted interference sum, improving SIR. It is not valid to assume that three-sector operation always divides the interferer count by exactly three. Actual antenna patterns, side-lobe levels, geometry and sector boundaries must be measured.

3.9 SIR under dynamic channel assignment

Dynamic assignment does not automatically change the physical propagation equation. Its controller chooses a channel only when the candidate channel satisfies the reuse and interference constraints. Thus, the physical SIR remains determined by wanted and interfering received powers. The benefit of DCA is improved utilisation of idle channels and a possible reduction in blocking, not an automatic increase in the total radio spectrum.

3.10 Cost–capacity relationship

Let C_survey be the one-time cost of traffic surveys and propagation measurements, C_site the cost of one installed site, C_sector the additional sector equipment cost per site, and C_control the controller or software cost. For M sites, a simplified cost model is:

C_total = C_survey + M·C_site + M·C_sector + C_control

If each cell supports A_B Erlangs, aggregate offered-traffic capacity is approximately M·A_B. Therefore:

Cost per supported Erlang = C_total/(M·A_B)

FCA has relatively low control complexity but may require more sites when traffic is uneven. Sectorisation adds antenna and radio costs but may improve SIR and capacity per site. DCA adds controller and coordination costs but can reduce unused capacity in lightly loaded cells. Exact currency values cannot be calculated because equipment prices, rent, power and survey rates are not supplied.

TASK 4 – ANALYSE BEST, AVERAGE AND WORST-CASE BEHAVIOUR

Radio performance varies with the mobile position inside a cell and with the traffic load at the time of observation. A user close to the serving base station generally receives a stronger wanted signal than a user at the cell edge. At the same time, the number and strength of visible co-channel interferers depend on geometry, antenna patterns and propagation conditions.

Case	Mobile position and traffic	SIR behaviour	Blocking behaviour
Best	Near serving base station; off-peak	Short wanted-signal distance gives high received power and generally high SIR	Few channels are occupied; blocking is close to zero
Average	Typical locations; average busy hour	Wanted and interfering powers are moderate	Blocking approaches the design target only near planned load
Worst	Cell edge or corner; event surge	Wanted signal is weak and multiple neighbours may interfere	Channels fill rapidly and blocking increases

4.1 Best-case analysis

Near the serving base station, the wanted signal distance d is smaller than the nominal cell-edge radius R. Under the path-loss model, wanted power is proportional to d^(−n). Since d < R, the wanted signal is stronger than the cell-edge reference. If the interference geometry remains approximately unchanged, SIR increases. During off-peak periods, fewer channels are occupied, so the probability that a new call finds all channels busy is low.

4.2 Average-case analysis

At ordinary user positions, the wanted signal is neither at its strongest nor at its weakest. The SIR depends on the user's distance, the directions of co-channel cells and local shadowing. During the average busy hour, channel occupancy increases. The Erlang-B blocking probability is a function of

both the offered traffic A and channel count m. It rises as offered traffic approaches the 14.036 Erlang planning limit for 21 channels.

4.3 Worst-case analysis

At a cell edge or corner, the mobile is farther from the serving base station and may be relatively close to more than one neighbouring cell. Wanted power falls and interference can become significant. If an event surge also raises offered traffic, all channels may become occupied more frequently. This creates both a radio-quality problem and a capacity problem.

4.4 Numerical illustration and shadowing

For N = 16 and n = 2.5, the six-equal-interferer approximation gives SIR ≈ 13.23 dB. This is only 1.23 dB above the 12 dB target. If shadowing reduces the wanted signal by 6 dB while interference remains unchanged, the effective SIR may fall to approximately 7.23 dB, below the target. This illustrates why a design with a small theoretical margin must be validated by field measurements and antenna optimisation.

A higher path-loss exponent reduces distant co-channel interference more rapidly and increases the ideal SIR estimate. However, a high exponent in a dense built-up area can also indicate stronger attenuation and penetration loss. Buildings may create shadowed areas and coverage holes. Therefore, a higher calculated SIR does not automatically guarantee better user experience.

The township-wide traffic totals are 25, 250, 500 and 2,500 Erlangs for the four subscriber levels. Blocking cannot be inferred from the subscriber total alone: it depends on the distribution of traffic among cells and the channels available in each cell. A stadium hotspot may be overloaded even when the township-wide average appears acceptable.

TASK 5 – SCALING AND SENSITIVITY ANALYSIS

Scaling analysis describes how network capacity, interference and site count change when cluster size, cell radius, subscriber population or propagation conditions change. It is important because the network must serve a growing population without receiving additional spectrum.

Change	Mathematical effect	Planning interpretation
N increases	Channels/cell ∝ 1/N; ideal SIR ∝ N^(n/2)	Better interference protection but fewer channels per cell
R decreases	Cell count ∝ $1/R^2$	More sites and higher aggregate capacity, but more handoffs and cost
Subscribers increase	Offered traffic ∝ subscribers	Capacity must rise through smaller cells, sectorisation and/or DCA
n increases from 2.5 to 4	Ideal SIR increases for fixed N	Distant interference falls faster, but coverage behind obstacles may worsen

5.1 Effect of cluster size

For fixed total spectrum S, channels per cell are S/N. Therefore, increasing N reduces the channel count available to each cell. However, the reuse ratio Q = √(3N) increases, and the ideal SIR expression is proportional to N^(n/2). The designer must therefore balance interference protection against traffic capacity.

5.2 Effect of cell radius

The hexagonal cell area is proportional to R^2. For a fixed township area, the number of cells is therefore proportional to $1/R^2$. Reducing the cell radius increases the number of cells and allows more spatial reuse. It also increases site acquisition, backhaul, power, maintenance and handoff activity.

5.3 Effect of subscriber growth

The offered traffic increases linearly with subscriber count under the fixed 0.025 Erlang per subscriber assumption. The 100,000-subscriber case requires approximately 179 cells at 14.036 Erlangs per cell. This is substantially higher than the 19-cell geometric coverage estimate for R = 0.5 km. Thus, coverage alone is not sufficient; the design must become capacity-driven.

5.4 Break-even point

A cell approaches its planned capacity limit when its offered traffic nears 14.036 Erlangs for 21 channels at a 2% blocking target. At or beyond this point, additional capacity is required through cell splitting, sectorisation or channel borrowing. Beyond 100,000 subscribers, a macro-only design is unlikely to be economical or adequate. A hierarchical macro–micro/small-cell arrangement is more practical because it preserves broad-area coverage while concentrating capacity in hotspots.

TASK 6 – FORMULATE THE RECURRENCE FOR REPEATED CELL SPLITTING

Cell splitting is a capacity-enhancement method in which an overloaded cell is replaced by smaller cells. In this assignment, the radius is reduced from R to R/2. Since cell area is proportional to R^2, each child cell covers one-quarter of the original area. One parent cell is therefore replaced by four child cells.

Let C_k be the number of cells after k uniform splitting rounds, K_k the ideal aggregate traffic capacity, R_k the cell radius, P_k the transmit power per cell and H_k the relative handoff rate. The base case is the original network at k = 0.

Quantity	Recurrence	Closed form
Cell count	C_k = 4C_(k−1)	C_k = C_0·4^k
Aggregate capacity	K_k = 4K_(k−1)	K_k = K_0·4^k
Cell radius	R_k = R_(k−1)/2	R_k = R_0/2^k
Transmit power	P_k = P_(k−1)/2^n	P_k = P_0/2^(nk)
Relative handoff rate	H_k ≈ 2H_(k−1)	H_k ≈ H_0·2^k

6.1 Explanation of the recurrence

The cell-count recurrence follows because every parent cell is replaced by four child cells. The same factor-of-four increase is used for ideal aggregate capacity under the simplifying assumption that each new cell provides a similar traffic-capacity unit and the traffic distribution remains suitable. The radius is halved at every round.

The transmit-power relation follows the planning model P′ = P/2^n. When the required radio distance is halved, the power needed to maintain a comparable received level falls according to the path-loss exponent. The relative handoff rate is approximated as doubling because smaller cells create more cell boundaries and more frequent transitions.

6.2 Selective splitting

Uniform splitting is not always economical because only some areas may be overloaded. Let f be the fraction of current cells selected for splitting. Each selected cell is replaced by four cells, adding three cells to the network for every selected parent. The recurrence becomes:

C_k = C_(k−1)(1 + 3f)

Under the simplified assumption that each split cell adds three equivalent capacity units:

K_k = K_(k−1)(1 + 3f)

If the split fraction changes by round, then C_k = C_0 × Π(1 + 3f_j). Selective splitting is more realistic for the township because academic blocks, hostels and stadium zones may require more capacity than residential areas.

6.3 Work and cost per round

Each splitting round requires site planning, installation, antenna configuration, frequency coordination, backhaul provisioning, neighbour-list updates and field testing. The work and cost increase with the number of cells. The design must also consider power limits, regulatory requirements, handoff performance and interference between the new small cells and the existing macro layer.

TASK 7 – SOLVE THE RECURRENCE AND INTERPRET THE RESULT

7.1 Uniform splitting by iteration

The uniform capacity recurrence is K_k = 4K_(k−1), with base case K_0. Expanding by substitution gives K_1 = 4K_0, K_2 = 4(4K_0) = 4²K_0, and K_3 = 4(4²K_0) = 4³K_0. Therefore, after k rounds:

K_k = K_0·4^k

To support a forecast demand D, the capacity must satisfy K_0·4^k ≥ D. Dividing by K_0 gives 4^k ≥ D/K_0. Taking logarithms:

$k \geq \log_4(D/K_0)$

Since the number of splitting rounds must be a whole number, the minimum is:

$k_{min} = \text{ceil}(\log_4(D/K_0))$

7.2 Numerical example

Assume one original cell supports K_0 = 14.036 Erlangs and the target demand is D = 200 Erlangs. Then:

$k_{min} = \text{ceil}(\log_4(200/14.036)) = \text{ceil}(1.91) = 2$ rounds

After two rounds, ideal capacity is $14.036 \times 4^2 = 14.036 \times 16 = 224.576$ Erlangs. The number of cells is multiplied by 16, the radius becomes R_0/4, transmit power per cell becomes P_0/4^n and relative handoff rate becomes approximately four times the original.

7.3 Selective splitting solution

For a constant split fraction f, the selective capacity recurrence is $K_k = K_0(1 + 3f)^k$. Requiring $K_k \geq D$ gives:

$k_{min} = \text{ceil}[\log(D/K_0) / \log(1 + 3f)]$

This expression applies when $0 < f \leq 1$ and the simplified capacity scaling assumption is reasonable. If f = 0, no capacity increase occurs through cell splitting.

7.4 Interpretation

The mathematical minimum number of rounds is not automatically the economically best solution. Splitting increases the number of sites, backhaul connections, handoff events, maintenance tasks and coordination requirements. A practical design should split only overloaded areas, use sectorisation where it provides meaningful isolation and use DCA where neighbouring cells have spare eligible channels.

TASK 8 – IMPLEMENT AND TEST THE PLANNING STRATEGIES

A reproducible analytical simulator can be implemented in Python. It calculates valid cluster sizes, channels per cell, approximate SIR, Erlang-B blocking and the number of cells needed for the offered traffic. The following program compares fixed reuse, a simplified sectorisation approximation and dynamic channel assignment. The sector and dynamic outputs are planning indicators rather than full radio-network simulation results.

8.1 Python implementation

```
import math

TOTAL_CHANNELS = 336
BLOCKING_TARGET = 0.02
TRAFFIC_PER_USER = 0.025
MIN_SIR_DB = 12.0

def valid_cluster_sizes(limit=30):
```

```
    values = set()
    for i in range(limit):
        for j in range(limit):
            n = i*i + i*j + j*j
            if 1 <= n <= limit:
                values.add(n)
    return sorted(values)

def sir_db(N, path_loss_n, interferers=6):
    sir = (3*N)**(path_loss_n/2) / interferers
    return 10 * math.log10(sir)

def erlang_b(channels, offered):
    b = 1.0
    for m in range(1, channels + 1):
        b = offered*b / (m + offered*b)
    return b

def max_erlangs(channels, target=BLOCKING_TARGET):
    low, high = 0.0, 1000.0
    for _ in range(100):
        mid = (low + high) / 2
        if erlang_b(channels, mid) <= target:
            low = mid
        else:
            high = mid
    return low

def evaluate_fixed(N, n=2.5):
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
    sir = sir_db(N, n)
    return channels, capacity, sir

def evaluate_sectorised(N, n=2.5, effective_interferers=2):
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
    sir = sir_db(N, n, effective_interferers)
    return channels, capacity, sir

def evaluate_dynamic(N, n=2.5):
    # Dynamic borrowing improves utilisation only when
    # neighbouring channels pass interference checks.
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
    sir = sir_db(N, n)
    return channels, capacity, sir

def main():
    print("Valid cluster sizes:", valid_cluster_sizes())
    for users in [1000, 10000, 20000, 100000]:
        traffic = users * TRAFFIC_PER_USER
        print("\nSubscribers:", users, "Traffic:", traffic)
        for N in [3, 4, 7, 12, 16]:
            m, cap, sir = evaluate_fixed(N, 2.5)
            cells = math.ceil(traffic / cap)
            print("N:", N, "channels/cell:", m,
                  "capacity/cell:", round(cap, 3),
                  "SIR dB:", round(sir, 2),
                  "cells:", cells)
```

```
    print("\nSectorised illustration:",
          evaluate_sectorised(16, 2.5, 2))
    print("Dynamic assignment baseline:",
          evaluate_dynamic(16, 2.5))

if __name__ == "__main__":
    main()
```

8.2 How to execute the program

1. Install Python 3 on the computer.
2. Open a code editor such as Visual Studio Code or create a file named cellular_planning.py.
3. Copy the complete program into the file and save it.
4. Open a terminal in the same folder.
5. Run the command: python cellular_planning.py.
6. Record the printed cluster sizes, channels per cell, Erlang capacity, SIR and estimated cell count.

8.3 Test matrix

Test dimension	Values to test
Cluster size N	3, 4, 7, 12; N = 16 as selected candidate
Path-loss exponent n	2.5, 3 and 4
Antenna layout	Unsectorised and sectorised
Subscriber population	1,000; 10,000; 20,000; 100,000
Strategies	Fixed reuse, sectorised reuse and dynamic assignment

The same user locations and random seed should be used when extending the program to a Monte-Carlo simulation. The full extension should generate random mobile positions in hexagonal cells, calculate path loss and log-normal shadowing, apply sector antenna gains, select eligible channels and record SIR for each trial. The current compact program is an analytical calculator and does not generate Monte-Carlo user-location results.

8.4 Sanity check

For N = 7 and n = 4, the six-equal-interferer approximation gives SIR = $(3\times7)^2/6 = 21^2/6 = 73.5$. In decibels, $10 \log_{10}(73.5) \approx 18.66$ dB, which is approximately 18.7 dB. This matches the expected sanity check.

TASK 9 – RECORD AND PRESENT EXPERIMENTAL PERFORMANCE

The simulator should record the same performance measurements for every strategy and traffic level. This allows the radio-quality and capacity limitations to be compared using a common basis. A result should not be considered reproducible unless the input assumptions and simulation settings are stated.

Metric	Meaning	Recommended presentation

Mean SIR	Average signal-to-interference ratio	Mean value and SIR cumulative distribution function (CDF)
5th-percentile SIR	SIR exceeded by 95% of users	Table and CDF; indicates edge-user performance
Outage probability	Fraction of users below 12 dB SIR	Percentage bar chart
Blocking probability	Blocked call attempts divided by total attempts	Blocking versus offered traffic graph
Channels per cell	Traffic channels assigned to a cell	Comparison table
Sites required	Cells/sites needed for offered load	Capacity versus subscriber graph
Runtime	Time required to complete the simulation	Reproducibility table

9.1 SIR distribution

Mean SIR alone can hide poor performance for users near cell boundaries. The 5th-percentile SIR is important because it describes a weak-signal portion of the user population. The outage probability is calculated as the percentage of users whose SIR is below 12 dB. A cumulative distribution function should be plotted to show the proportion of users at or below each SIR value.

9.2 Blocking probability

Blocking probability should be plotted against offered traffic for each strategy. The 2% target should be shown as the design threshold. A strategy that maintains a lower blocking probability under uneven traffic may be preferable even if its theoretical SIR is not the highest.

9.3 Reproducibility settings

- Random seed and number of user locations.
- Number of call attempts and call-arrival model.
- Path-loss model and path-loss exponent.
- Shadowing standard deviation and penetration-loss assumptions.
- Antenna pattern, azimuth, tilt and sector configuration.
- Channel-allocation and borrowing rules.
- Cell layout, boundary conditions and simulation runtime.

The source assignment provides analytical planning values and recommended measurements, but it does not provide measured Monte-Carlo outputs. Therefore, numerical mean SIR, 5th-percentile SIR, outage percentage and runtime must be obtained by executing a fully specified simulation. They should not be invented or presented as measured results.

TASK 10 – COMPARE THEORY WITH SIMULATION

Analytical models provide a simplified prediction of network behaviour, while simulation represents more realistic geometry, user positions and radio conditions. The two methods should show the same overall trends even when their exact numerical values differ.

10.1 Expected agreement

- Larger cluster size generally increases co-channel separation and improves the ideal SIR.
- Larger cluster size reduces the number of channels assigned to each cell.
- Smaller cells increase site count and can increase aggregate capacity through spatial reuse.
- Sectorisation can reduce dominant interference when antenna directivity is effective.
- Dynamic assignment can reduce blocking when neighbouring cells have spare eligible channels.

10.2 Sources of deviation

- The analytical six-interferer model assumes six equal-distance interferers, whereas real cells have different distances and received powers.
- Shadowing and building penetration create random changes in signal strength.
- Finite network edges have fewer interfering neighbours than an infinite repeating grid.
- Users are not uniformly distributed; lecture halls, hostels and stadiums create traffic hotspots.
- Erlang-B assumes a loss system with independent arrivals and no waiting queue; real call and data traffic may differ.
- Integer rounding means that 336/N may not be an integer for every cluster size.
- Real sector antennas have side lobes and imperfect front-to-back isolation.
- Mobility and handoff behaviour may change the resource availability seen by a user.

10.3 Strategy comparison

FCA is suitable as a stable baseline because its channel allocation and interference pattern are predictable. Sectorisation is useful where directional traffic and antenna isolation can reduce dominant interferers. DCA is useful where neighbouring cells have uneven offered traffic and spare channels can be borrowed safely. A hybrid network can combine these advantages.

The best strategy must be selected using blocking, 5th-percentile SIR, outage, site count, runtime and total cost together. The strategy should not be declared best solely because it has the highest theoretical capacity. Simulation results should be compared with analytical predictions after the actual experiment has been run.

TASK 11 – RECOMMEND AND DEFEND THE FINAL DESIGN

11.1 Recommended network architecture

The recommended baseline is a hierarchical cellular network. Fixed frequency reuse provides the stable macro-layer frequency plan. Sectorisation is added in high-demand zones, and dynamic channel assignment or borrowing is used where neighbouring cells have uneven traffic. Microcells or

small cells are introduced as demand grows, particularly around hostels, academic blocks, commercial areas and the stadium.

11.2 Selected planning parameters

Parameter	Recommended baseline	Reason
Cluster size	N = 16	Conservative reuse candidate for the 12 dB SIR target
Total channels	336	Fixed licensed allocation
Channels per cell	21	336/16
Blocking target	≤ 2%	Specified grade of service
Traffic capacity per cell	Approximately 14.036 Erlangs	Erlang-B limit for 21 channels at 2% blocking
Minimum SIR	12 dB	Selected digital-style threshold
Path-loss planning case	n = 2.5 as conservative open-campus case	Produces the lowest of the three example SIR estimates

11.3 Expected capacity and interference

For N = 16, the approximate SIR values are 13.23 dB at n = 2.5, 17.44 dB at n = 3 and 25.84 dB at n = 4. The open-campus case has a modest margin above the 12 dB threshold. The design therefore requires careful antenna optimisation, site placement and field validation.

With 21 channels per cell, the approximate offered-traffic limit is 14.036 Erlangs at 2% blocking. For 100,000 subscribers, the total offered traffic is 2,500 Erlangs and the first-order capacity estimate requires 179 cells. The equal-hexagon radius for 179 cells covering 12 km² is approximately 0.161 km.

11.4 Cost and operational trade-offs

The macro layer provides broad coverage and avoids excessive site deployment in low-demand areas. Small cells add capacity in concentrated zones but increase site rent, power, backhaul and maintenance. Sectorisation improves spatial filtering but adds antenna and radio equipment. Dynamic borrowing improves trunking efficiency but requires reliable load reporting and interference coordination. The operator must compare the cost per supported Erlang and the quality of service achieved.

11.5 Growth beyond 100,000 subscribers

A single macro-cell layer is not sufficient for the 100,000-subscriber planning case under the stated traffic assumptions. The network must continue densification and capacity optimisation. A hierarchical macro–micro/small-cell layout is more practical because macro sites maintain broad coverage while smaller cells serve local hotspots. The design remains feasible only if the operator continues to add capacity and coordinate interference as demand grows.

11.6 Final defence

The hybrid recommendation is stronger than relying on only one strategy. FCA gives a predictable baseline, sectorisation improves spatial isolation in busy zones, and DCA uses eligible idle channels where traffic is uneven. Cell splitting provides additional capacity when a cell approaches its Erlang-B limit. The combination addresses interference, blocking, mobility and cost rather than maximising only one metric.

TASK 12 – DAY-4 DESIGN CHALLENGE: RE-EVALUATE AND ADAPT THE DESIGN

12.A Impact analysis

The changed scenario introduces a convocation or sports event with 25,000 attendees in a 1 km² stadium zone. Each attendee generates 0.05 Erlang, and the event lasts four hours. Events occur approximately twice per month. The event traffic is:

$A_{event} = 25{,}000 \times 0.05 = 1{,}250$ Erlangs

The event traffic density is 1,250 Erlangs/km². For comparison, the 100,000-subscriber township-wide traffic is 2,500 Erlangs over 12 km², giving an average density of 208.33 Erlangs/km². The event zone therefore has approximately 1,250/208.33 = 6 times the township-wide average traffic density.

The assumptions of uniform traffic, steady demand and permanent capacity no longer hold. The surge is concentrated in a small area and occurs for a limited duration. Since the event is predictable and occurs only about twice per month, temporary capacity is likely to be more economical than permanently redesigning the entire township.

12.B Evaluation of the original design

The original N = 16 design assigns 21 channels to each cell and supports approximately 14.036 Erlangs per cell at a 2% blocking target. The number of cell-equivalents required for the event is:

Required cell-equivalents = ceil(1,250/14.036) = 90

The original macro layout cannot safely serve this concentrated demand without additional capacity. Channel exhaustion and increased blocking are likely to occur first in the stadium cells. High user density can also increase interference, signalling load and handoff activity.

Township-wide spare capacity cannot automatically solve the stadium hotspot because spare channels can only be borrowed when they are available and satisfy interference and reuse constraints. Dynamic borrowing may help, but it cannot be assumed to provide the full 1,250 Erlangs.

12.C Adaptation or redesign

Decision: MODIFY TEMPORARILY. The normal township macro network should be retained, while temporary capacity is deployed around the stadium for event days. Temporary microcells or

cells-on-wheels (COWs), sectorised antennas and coordinated dynamic channel borrowing should be used.

- Pre-plan temporary frequency groups, backhaul, power supply, site permissions and handoff boundaries.
- Use approximately 90 cell-equivalents as a first-order minimum based on 14.036 Erlangs per cell; provide additional headroom for bursty demand and uncertainty.
- Divide the 1 km² stadium zone into 90 equal hexagonal cell areas for an illustrative radius estimate.
- Use 60-degree or 120-degree sectorisation where directional traffic flows justify it.
- Adjust antenna tilt and transmit power to reduce overshooting and interference with the macro layer.
- Use dynamic borrowing only after checking co-channel interference and reserving resources for handoffs and control signalling.
- After the event, switch off or redeploy temporary cells to avoid unnecessary permanent operating costs.

For 90 equal hexagonal cell areas in 1 km², the illustrative radius is:

$R \approx \sqrt{[1/(90 \times 2.598)]} \approx 0.065$ km ≈ 65 metres

This is a theoretical density estimate, not a final site map. Real deployment depends on stadium geometry, audience distribution, antenna mounting locations, safety requirements and backhaul access.

12.D Evidence-based defence

Temporary modification is preferred because the event is predictable, restricted to 1 km² and lasts only four hours, while event days occur roughly twice monthly. Permanent stadium-scale densification would incur site rent, power, maintenance and backhaul costs even when the stadium is not busy. Temporary COWs and microcells concentrate expenditure where and when it is required.

The traffic calculation supports the decision: 1,250 Erlangs is far above the capacity of a small number of macro cells and requires roughly 90 cell-equivalents at the selected per-cell Erlang-B limit. Sectorisation and dynamic borrowing can improve utilisation, but the operator must still provide sufficient radio resources and backhaul. Final acceptance should require measured SIR, outage and blocking during a controlled event trial.

12.E Risks and operational controls

Temporary cells must be integrated carefully with the permanent macro network. Poorly planned transmit power can create overshooting and additional co-channel interference. Incorrect neighbour lists can increase dropped calls and handoff failures. Backhaul congestion can limit user throughput even when radio capacity is sufficient. The operator should conduct pre-event testing, monitor performance during the event and remove or redeploy temporary equipment after the event.

Issue	Control measure

High call blocking	Add temporary radio capacity and maintain spare channel headroom
Co-channel interference	Coordinate reuse pattern, power and antenna direction
Handoff failures	Prepare neighbour lists and test mobility paths
Backhaul congestion	Provision adequate temporary backhaul capacity
Temporary-site cost	Deploy only for event periods and redeploy equipment afterwards

OVERALL CONCLUSION

The assignment demonstrates that cellular network planning is a balance between spectrum reuse, interference, traffic capacity, coverage, cost and mobility. Fixed cluster frequency reuse provides a predictable baseline. Sectorisation improves spatial filtering by reducing the impact of unwanted signals. Dynamic channel assignment improves channel utilisation when traffic varies across neighbouring cells. Cell splitting and hierarchical deployment provide additional capacity as the subscriber population grows.

For the given 336-channel system, N = 16 is selected as a conservative planning candidate. It provides 21 channels per cell and approximately 14.036 Erlangs per cell at a 2% blocking target under Erlang-B. The ideal six-interferer SIR estimates are 13.23 dB, 17.44 dB and 25.84 dB for path-loss exponents 2.5, 3 and 4 respectively. The open-campus estimate has only a modest margin above the selected 12 dB threshold, making field validation essential.

As the subscriber population increases, the network must add smaller cells and use a hierarchical macro–micro/small-cell layout. For 100,000 subscribers, the first-order estimate requires approximately 179 cells at the selected per-cell traffic limit. A macro-only network is therefore insufficient for the full planning horizon. The recommended design combines fixed reuse, sectorisation, dynamic channel borrowing and selective cell splitting.

For the four-hour stadium event, the offered traffic is 1,250 Erlangs in 1 km², approximately six times the township-wide average traffic density. The preferred response is temporary modification using microcells or cells-on-wheels, sectorised antennas and carefully controlled channel borrowing. This provides event capacity without imposing the ongoing cost of permanent stadium-scale densification.

The final network should be accepted only after measurements confirm the required SIR, outage, blocking and handoff performance under realistic user distributions. The mathematical models provide a sound planning foundation, while field surveys and reproducible simulation are necessary to validate the deployed network.

REFERENCES

- Mobile Computing – ITA0329 Assignment-1 (CO1), supplied assignment brief and planning constraints.
- Mobile Computing – Assignment-1 reference solution supplied for formatting and analytical guidance.

- Standard cellular network planning concepts: hexagonal frequency reuse, Erlang-B traffic modelling, co-channel interference, sectorisation and dynamic channel assignment.
- Python 3 documentation for implementation of mathematical calculations and iterative procedures.